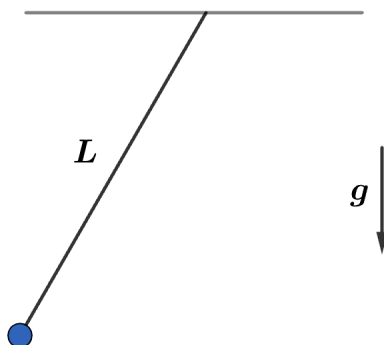


2018A F=ma Exam: Problem 12

Kevin S. Huang



Theory:

$$T = 2\pi\sqrt{\frac{L}{g}}$$

We have

$$L = 1.00 \pm 0.05 \text{ m}$$

$$T = 2.00 \pm 0.10 \text{ s}$$

Recall when two uncorrelated quantities are multiplied, the relative (fractional) uncertainties add in quadrature. Also recall the relative certainty gets multiplied by $|n|$ when the quantity is raised to the n th power. Thus,

$$g = L \left(\frac{2\pi}{T} \right)^2 = 9.9 \text{ m/s}^2$$

$$\frac{\delta g}{g} = \sqrt{\left(\frac{\delta L}{L} \right)^2 + \left(\frac{2\delta T}{T} \right)^2}$$

$$\delta g = 9.9 \sqrt{\left(\frac{0.05}{1} \right)^2 + \left(\frac{2(0.10)}{2.00} \right)^2} = 1.1 \text{ m/s}^2$$

so the answer is D.